

New Year Plan

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Outline

- What I did since last lab meeting (25/11/20)
- What I'm doing now
 - Projects
 - Progress Update
- Dynamic Decoupling
 - CPMG Sequence
 - Finding τ Analytically
 - Conditional Operation
 - CPMG analysis
- DDrf
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 - DDrf: Analytical Approach
 - DDrf Spectroscopy
- Slack + Notion

What I did since last lab meeting (25/11/20)

- Assignments (~2 weeks)
- Grad Admission (25/12/17)
- Final Exam (~1 week) → Computer Network (A), Programming Language (A+)
- Wrote abstract for conference (26/01/07)
- Interview from UC Davis (26/01/14)
 - If accepted, Open house (26/02/22)
- And many other personal stuffs ...

Projects

1. [**Ongoing**] Hybrid DDrf: Enhanced Electron Nuclear control leveraging hyperfine coupling and radio-frequency drive
2. [Holding] Observable Expectation Value Extraction (with Geonhee Kim)
3. [Holding] Entanglement Generation / Quantification + AI? (with Seongpyo Hong)
4. [Pending] Compressive QST
5. [Ongoing] Slack + Notion

Progress Update

Completed Tasks

- Analytical CPMG study: Effect of τ_{MW}
 - Not that meaningful result though ...

In Progress

- DDrf Spectroscopy
- Hybrid DDrf conditional gate strategy
- Slack + Notion

Pending

- Observable Expectation Value Extraction (Theoretical background work completed)
- Entanglement Generation / Quantification
- Compressive QST

Dynamic Decoupling

Original Purpose: Extending Coherence Time

CPMG was originally used to increase coherence time, but can also be used for various purposes, including implementing conditional gates.

Hamiltonian

$$\mathcal{H} = \underbrace{\gamma_c B_z I_z^i + A_{||}^i S_z I_z^i + A_{\perp} S_z I_x^i}_{\text{Drift Hamiltonian}} + \underbrace{\frac{1}{\sqrt{2}} \Omega_{\text{MW}} S_x}_{\pi\text{-pulse}}$$

$$\rightarrow \mathcal{H} = |0\rangle\langle 0| \otimes \omega_0 I_z + |-1\rangle\langle -1| \otimes (\omega_L I_z - A_{||} I_z - A_{\perp} I_x)$$

$$\rightarrow \mathcal{H} = |0\rangle\langle 0| \otimes H_0 + |-1\rangle\langle -1| \otimes H_1$$

That is, nuclear spin evolves by $e^{-iH_{0(1)}t}$ when electron spin is in state $|0\rangle(|1\rangle)$.

Figure. Diagram for CPMG sequence/ CPMG is a repetition of $(\tau - \pi - 2\tau - \pi - \tau)$ for $\frac{N}{2}$ times. The figure is extracted from [Phys. Rev. X 15, 021011](#).

CPMG Sequence

- Initial electron spin state 0(1): $e^{-i\phi\hat{I}\cdot\hat{\sigma}^i}$
 - if NV spin is 0-state: $e^{-i\phi\hat{I}\cdot\hat{\sigma}^0} = e^{-iH_0\tau}e^{-iH_12\tau}e^{-iH_0\tau}$
 - if NV spin is 1-state: $e^{-i\phi\hat{I}\cdot\hat{\sigma}^1} = e^{-iH_1\tau}e^{-iH_02\tau}e^{-iH_1\tau}$
- When vectors σ^0 and σ^1 are anti-parallel, a conditional gate is achieved
- The directions are generally parallel, but become anti-parallel at specific τ values

Note: ϕ is the same regardless of whether the initial state is 0 or 1 in the CPMG train.

Finding τ Analytically

To enable a conditional gate via repetition of CPMG train, we should choose τ such that $\hat{\sigma}^0 \cdot \hat{\sigma}^1 = -1$. τ can be found analytically under a strong magnetic field. [\[Phys. Rev. Lett. 109, 137602\]](#)

$$\tau \simeq \frac{(2k-1)\pi}{2\gamma_c B_z + A_{||}}$$

Then, the number of repetitions of CPMG train is chosen meticulously to achieve $N\phi = \frac{\pi}{2}$.

Conditional Operation

At a certain τ , we observe a conditional operation. Red (Blue) arrow denotes the rotation axis when the initial electron spin state was $|0\rangle$ ($|1\rangle$).

CPMG analysis

- Initial electron spin state 0(1): $e^{-i\phi\hat{I}\cdot\hat{\sigma}^i}$
 - if NV spin is 0-state: $U_0 = e^{-i\phi\hat{I}\cdot\hat{\sigma}^0} = e^{-iH_0\tau}e^{-iH_12\tau}e^{-iH_0\tau}$
 - if NV spin is 1-state: $U_1 = e^{-i\phi\hat{I}\cdot\hat{\sigma}^1} = e^{-iH_1\tau}e^{-iH_02\tau}e^{-iH_1\tau}$

If one CPMG train can be described this way, the unitary gate takes the form of a block matrix:

$$U^{\frac{N}{2}} = \begin{pmatrix} U_0^{\frac{N}{2}} & 0 \\ 0 & U_1^{\frac{N}{2}} \end{pmatrix}$$

where off-diagonal components are zero.

However, if the time for the MW pulse implementing the π -pulse gate is not short enough, it can cause additional effects beyond those described below.

CPMG analysis

However, if we calculate U with non-negligible MW time, off-diagonal components become non-zero and each block matrix is no longer unitary.

$$U^{\otimes \frac{N}{2}} = \left(e^{-i\tau H_0} e^{-i\tau_{\text{MW}}(H_0 + H_{\text{MW}})} e^{-i2\tau H_0} e^{-i\tau_{\text{MW}}(H_0 + H_{\text{MW}})} e^{-i\tau H_0} \right)^{\frac{N}{2}} = \begin{pmatrix} V_{00} & V_{01} \\ V_{10} & V_{11} \end{pmatrix}$$

Its operation on a state is described as:

$$|\psi\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \otimes |\phi\rangle$$

then, we have

$$|\psi\rangle \rightarrow \frac{1}{\sqrt{2}}(|0\rangle \otimes \underbrace{(V_{00} + V_{01})}_{=V^0}|\phi\rangle + |1\rangle \otimes \underbrace{(V_{11} + V_{10})}_{=V^1}|\phi\rangle)$$

CPMG analysis

Both V_0 and V_1 are not unitary. To analyze them, I used polar decomposition which is related to SVD.

$$V = UP = (uv^\dagger)(v\Sigma v^\dagger)$$

where u, v are unitary matrices derived from SVD. Here U is unitary, while P is non-unitary and can be regarded as a quantum channel.

$$P = \sum_i \lambda_i |\lambda_i\rangle \langle \lambda_i|$$

where λ_i are the singular values. Thus, we have

$$|\psi\rangle \rightarrow \frac{1}{\sqrt{2}} (|0\rangle \otimes U_0 P_0 |\phi\rangle + |1\rangle \otimes U_1 P_1 |\phi\rangle)$$

CPMG analysis

We can analyze from two perspectives:

- How $\vec{\sigma}_0 \cdot \vec{\sigma}_1$ (with respect to U_0 and U_1) changes with t_{MW} and N .
- How the singular values $\lambda_{0,1}$ change with t_{MW} and N .

For this tentative simulation, I used the following parameters:

Param	Value	Description	Param	Value	Description
B_z	440.1 G	Magnetic field strength	τ	0.1 - 20 μs	Free evolution time (varied)
$A_{\parallel}$	(2π) 130 kHz	Parallel hyperfine coupling	τ_{MW}	0.1 - 50 ns	π -pulse duration (varied)
$A_{\perp}$	(2π) 50 kHz	Perpendicular hyperfine coupling	N	2 - 40	Number of CPMG repetitions

CPMG analysis: $\vec{\sigma}_0 \cdot \vec{\sigma}_1$

The following colormap shows how the $\vec{\sigma}_0 \cdot \vec{\sigma}_1$ value changes with τ and τ_{MW} .

τ shifts to shorter values as τ_{MW} increases. However, the amount of this shift keeps the total time for the CPMG train constant.

CPMG analysis: $\vec{\sigma}_0 \cdot \vec{\sigma}_1$

When varying N while fixing τ_{MW} , the value of τ that makes the total operation conditional, $\vec{\sigma}_0 \cdot \vec{\sigma}_1 = -1$, remains unchanged.

CPMG analysis: λ_i

If $\lambda_i \simeq 1$, P can be approximated as the identity matrix.

$$P = \sum_i \lambda_i |\lambda_i\rangle \langle \lambda_i|$$

This operator amplifies or attenuates components based on λ_i . While the overall operation preserves norm, P alone does not preserve the norm of the state vector.

Quick example:

Suppose $\lambda_i = \left\{ \sqrt{\frac{3}{2}}, \sqrt{\frac{1}{2}} \right\}$, with $|\lambda_0\rangle = |0\rangle$ and $|\lambda_1\rangle = |1\rangle$.

If the prepared state is $|+\rangle$, it transforms to

$$|+\rangle \rightarrow \frac{\sqrt{3}}{2} |0\rangle + \frac{1}{2} |1\rangle$$

CPMG analysis: λ_i

As expected, when τ_{MW} and N increase, the singular value λ_i deviates from 1.

[Update] CPMG analysis: λ_i

In a CPMG sequence, we perform an X -basis measurement at the end of the sequence. Without considering the drift Hamiltonian during π -pulse time, we have

$$\begin{aligned} P_x &= \text{Tr}(|+\rangle\langle+| \otimes I) \rho = \text{Tr}(|+\rangle\langle+| \otimes I) U \rho_0 U^\dagger \\ &= \frac{1}{4} \text{Tr} U_0 \rho_{^{13}\text{C}} U_0^\dagger + \frac{1}{4} \text{Tr} U_0 \rho_{^{13}\text{C}} U_1^\dagger + \frac{1}{4} \text{Tr} U_1 \rho_{^{13}\text{C}} U_0^\dagger + \frac{1}{4} \text{Tr} U_1 \rho_{^{13}\text{C}} U_1^\dagger \\ &= \frac{1}{2} \left(\frac{1}{2} \Re(\text{Tr} U_0 U_1^\dagger) + 1 \right) \end{aligned}$$

where

$$\rho_0 = |+\rangle\langle+| \otimes \rho_{^{13}\text{C}}$$

and the last line follows from $\rho_{^{13}\text{C}} = I$.

[Update] CPMG analysis: λ_i

Now it becomes more complex...

$$\begin{aligned} P_x &= \text{Tr}(|+\rangle\langle+| \otimes I) \rho = \text{Tr}(|+\rangle\langle+| \otimes I) U \rho_0 U^\dagger \\ &= \text{Tr} \frac{1}{4} \begin{pmatrix} I & I \\ I & I \end{pmatrix} \begin{pmatrix} V_{00} & V_{01} \\ V_{10} & V_{11} \end{pmatrix} \begin{pmatrix} \rho & \rho \\ \rho & \rho \end{pmatrix} \begin{pmatrix} V_{00}^\dagger & V_{10}^\dagger \\ V_{01}^\dagger & V_{11}^\dagger \end{pmatrix} \\ &= \frac{1}{4} \text{Tr}(\sigma_{00} + \sigma_{01} + \sigma_{10} + \sigma_{11}) \end{aligned}$$

where

$$\begin{aligned} \sigma_{00} &= (V_{00} + V_{01}) \rho^{13}\text{C} (V_{00}^\dagger + V_{01}^\dagger) &= U_0 P_0 \rho^{13}\text{C} P_0 U_0^\dagger \\ \sigma_{01} &= (V_{00} + V_{01}) \rho^{13}\text{C} (V_{10}^\dagger + V_{11}^\dagger) &= U_0 P_0 \rho^{13}\text{C} P_1 U_1^\dagger \\ \sigma_{10} &= (V_{10} + V_{11}) \rho^{13}\text{C} (V_{00}^\dagger + V_{01}^\dagger) &= U_1 P_1 \rho^{13}\text{C} P_0 U_0^\dagger \\ \sigma_{11} &= (V_{10} + V_{11}) \rho^{13}\text{C} (V_{10}^\dagger + V_{11}^\dagger) &= U_1 P_1 \rho^{13}\text{C} P_1 U_1^\dagger \end{aligned}$$

[Update] CPMG analysis: λ_i

However, its effect does not seem to be critical.

$$\begin{aligned}\sigma_{00} + \sigma_{11} &\simeq 2 \\ \sigma_{01} + \sigma_{10} &= \Re(\text{Tr} V_0 V_1^\dagger)\end{aligned}$$

Then,

$$\Re(\text{Tr} V_0 V_1^\dagger) = 2p_0 p_1 (\cos^2 \phi + \sin^2 \phi (\hat{\sigma}_0 \cdot \hat{\sigma}_1)) (1 + r_0 r_1 (\hat{\mu}_0 \cdot \hat{\mu}_1))$$

where p_0, p_1 are average of eigenvalues, ϕ is conditional rotation angle, $\hat{\sigma}_i$ is rotation-axis vector related to U_i , and μ_i is the vector related to quantum channel P_i . So $r_0 r_1$ term is responsible for the non-negligible τ_{MW} , but its order is at most $\mathcal{O}(10^{-2})$.

CPMG Fitting

CPMG Fitting

We measure the probability, P_x , that the initial state of electron spin is preserved.

$$M = \Re(\text{Tr } U_0 U_1^\dagger) = 1 - (1 - \hat{n}_0 \cdot \hat{n}_1) \sin^2 \frac{N\phi}{2}$$
$$P_x = \frac{M e^{-\frac{2N\tau}{T_\alpha}}}{2} + \frac{1}{3} + \frac{e^{-\frac{2N\tau}{T_\beta}}}{6}$$
$$\cos \phi = \cos \alpha \cos \beta - \cos \beta \sin \alpha \sin \beta$$
$$1 - \hat{n}_0 \cdot \hat{n}_1 = \sin^2 \beta \frac{(1 - \cos \alpha)(1 - \cos \beta)}{1 + \cos \phi}$$

where

$$\cos \beta = \frac{\omega_L - \omega_{\parallel}}{\sqrt{(\omega_L - \omega_{\parallel})^2 + A_{\perp}^2}}$$
$$\sin \beta = \frac{A_{\perp}}{\sqrt{(\omega_L - \omega_{\parallel})^2 + A_{\perp}^2}}$$
$$\alpha = \tau \sqrt{(\omega_L - \omega_{\parallel})^2 + A_{\perp}^2}$$
$$\beta = \omega_L \tau$$

CPMG Fitting

Strategy

1. Using three values of N for fitting data, while the remaining are used for validation data.
2. The cost function is based on Mean Squared Error (MSE). The cost function can be adjusted so that peak data could be regarded more crucially. For example, $\frac{(y_{\text{data}} - y_{\text{fit}})^2}{y_{\text{data}}}$, where $0 < y_{\text{data}} < 1$.
3. By assuming the number of ^{13}C spins, perform fitting and check validation. It was expected that training error would decrease while validation error increases. In this process, one might choose the desired number of ^{13}C spins.

Problem → Failed!

However, every time I performed fitting, I obtained different hyperfine coupling parameters.

DDrf Milestone

1. DDrf Spectroscopy (Experiment) <- (Eunsang Lee/ Donghun Jung)
→ Output: ω_1 of ^{13}C spins
2. DDrf Spectroscopy (Experiment)
→ Output: $\sin \beta$, $A_{\perp}$
3. DDrf/DD comparative study (Theory)
4. Hybrid DDrf Gate (Theory) <- (Jiwon Jeon/ Donghun Jung)
→ Further faster conditional gate implementation

DDrf: Analytical Approach

As the nuclear spin quantization axis is dependent on the electron spin state, the Hamiltonian becomes

$$\begin{aligned} H &= |0\rangle\langle 0| \otimes H_0 + |1\rangle\langle 1| \otimes H_1 \\ H_0 &= \omega_0 I_z + 2\Omega_{\text{RF}} \cos(\omega_{\text{RF}} t + \phi) I_x \\ H_1 &= \omega_1 \tilde{I}_z + 2\Omega_{\text{RF}} \cos \beta \cos(\omega_{\text{RF}} t + \phi) \tilde{I}_x + 2\Omega_{\text{RF}} \sin \beta \cos(\omega_{\text{RF}} t + \phi) \tilde{I}_z \end{aligned}$$

where

$$\begin{aligned} \cos \beta &= \frac{\omega_0 - A_{\perp}}{\omega_1} & \tilde{I}_z &= \cos \beta I_z + \sin \beta I_x \\ \sin \beta &= \frac{A_{\perp}}{\omega_1} & \tilde{I}_x &= \cos \beta I_x - \sin \beta I_z. \end{aligned}$$

DDrf: Analytical Approach

Two rotating frames are used with respect to electron spin state;

$$R_0(t) = e^{-i\omega_{\text{RF}}tI_z} \quad R_1(t) = e^{-i\omega_{\text{RF}}t\tilde{I}_z}$$

In rotating frame, each Hamiltonian becomes:

$$\begin{aligned} H_0 \rightarrow H'_0 &= R_0(t)(H_0 - \omega_{\text{RF}}I_z)R_0(t)^\dagger \\ &= (\omega_0 - \omega_{\text{RF}})I_z + \Omega_{\text{RF}}(\cos \phi I_x + \sin \phi I_y) \end{aligned}$$

$$\begin{aligned} H_1 \rightarrow H'_1 &= R_1(t)(H_1 - \omega_{\text{RF}}I_z)R_1(t)^\dagger \\ &= (\omega_1 - \omega_{\text{RF}})\tilde{I}_z + \Omega_{\text{RF}} \cos \beta (\cos \phi \tilde{I}_x + \sin \phi \tilde{I}_y) \\ &= (\omega_1 - \omega_{\text{RF}})(\cos \beta I_z + \sin \beta I_x) + \Omega_{\text{RF}} \cos \beta (\cos \beta \cos \phi I_x + \sin \phi I_y - \sin \beta \cos \phi I_z) \end{aligned}$$

In each rotating frame, time evolution can be calculated readily; $e^{-iH_{0(1)}t}$.

DDrf: Analytical Approach

Then, throughout full time evolution, the unitary operation can be calculated as:

$$\begin{aligned} U &= |0\rangle\langle 0| \otimes U_0 + |1\rangle\langle 1| \otimes U_1 \\ U_0 &= R_0(4N\tau)^\dagger e^{-iH'_0\tau} R_0((2N-1)\tau) R_1((2N-1)\tau)^\dagger e^{-iH'_1 2\tau} R_1((2N-3)\tau) R_0((2N-3)\tau)^\dagger e^{-iH'_0\tau} \dots \\ &\quad \dots e^{-iH'_0\tau} R_0(3\tau) R_1(3\tau)^\dagger e^{-iH'_1 2\tau} R_1(\tau) R_0(\tau)^\dagger e^{-iH'_0\tau} R_0(0) \\ U_1 &= R_1(4N\tau)^\dagger e^{-iH'_1\tau} R_1((2N-1)\tau) R_0((2N-1)\tau)^\dagger e^{-iH'_0 2\tau} R_0((2N-3)\tau) R_1((2N-3)\tau)^\dagger e^{-iH'_1\tau} \dots \\ &\quad \dots e^{-iH'_1\tau} R_1(3\tau) R_0(3\tau)^\dagger e^{-iH'_0 2\tau} R_0(\tau) R_1(\tau)^\dagger e^{-iH'_1\tau} R_1(0) \end{aligned}$$

It is worth mentioning that

- $\Omega_{\text{RF}} \rightarrow 0, \omega_{\text{RF}} = \omega_1$: CPMG at a certain $\tau \simeq \frac{(2k-1)\pi}{2\omega_0 + A_{\parallel}}$
- $\beta \rightarrow 0, \omega_{\text{RF}} = \omega_1 > \omega_0$: DDrf(2019)
- $\Omega_{\text{RF}} \rightarrow \frac{\Omega_{\text{RF}}}{\cos \beta}$: Jiwon's idea
- $\tau = \frac{(2k-1)\pi}{2\omega_0 + A_{\parallel}}$: Hybrid DDrf

DDrf Spectroscopy

The paper, [Physical Review X 9.3 \(2019\): 031045](#), suggested that the DDrf gate provides the additional benefit that spins with small $A_{\perp}$ can also be detected.

Procedure:

1. $\frac{\pi}{2}$ -pulse rotates electron spin to $|+\rangle$.
2. DDrf Gate with fixed N and τ .
3. $\frac{\pi}{2}$ -pulse is applied to electron spin with varying phase ϕ .

Then, at the resonant frequency, peaks are observed.

$$\omega_{\text{RF}} = \omega_1 + \frac{2\pi m}{\tau}$$

where m is an integer.

DDrf Spectroscopy



DDrf Spectroscopy

Problems

1. Peaks were not observed for all resonant frequencies.
2. The peaks corresponding to ω_{+1} and ω_{-1} must be different.

Potential Issues

1. Incorrect coding.
2. Additional implementation required for sweeping phase ϕ of the electron spin.

Slack + Notion

Problems

1. Data stored in Slack chat is not retained for more than 90 days.
2. As we are not using Notion or other tools, the management of lab resources is contingent on each individual's ability.

→ We have craved a systemic and integrated management system.

Based on the following clone coding, I'm working on building a Slack alternative and integrating Notion features. We will potentially be able to record lab notes here. Of course, before this, we must reach an agreement on writing lab notes.

Slack + Notion

Future Work

1. [**Ongoing**] Hybrid DDrf: Enhanced Electron Nuclear control leveraging hyperfine coupling and radio-frequency drive
2. [Holding] Observable Expectation Value Extraction (with Geonhee Kim)
3. [Holding] Entanglement Generation / Quantification + AI? (with Seongpyo Hong)
4. [Pending] Compressive QST
5. [Ongoing] Slack + Notion

I'm going to be a "FAT" person!